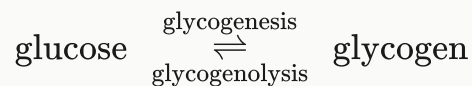


Liver Metabolic Role

You already know the liver "processes what you eat" — but the deeper story is that it's a **chemical distribution center** sitting between your gut and the rest of your body, deciding what gets used now, stored for later, or thrown out.

After digestion, blood from your intestines flows through the *hepatic portal vein* straight into the liver, carrying glucose, amino acids, and fats. The liver's first job is buffering: it doesn't let your blood sugar spike or crash. When glucose is abundant, liver cells convert it into a branched storage molecule called **glycogen**, via a process called glycogenesis. When blood sugar drops — say, between meals — the liver reverses the reaction (glycogenolysis), releasing glucose back into the blood. This is a feedback loop:



The liver also manufactures **bile**, a fluid stored in the gallbladder that emulsifies fats in the small intestine — breaking large fat droplets into smaller ones so digestive enzymes can reach more surface area. Meanwhile, the liver's detox function runs largely through two enzyme phases: Phase I reactions (often oxidation) make toxins more reactive, and Phase II reactions attach a molecule to make them water-soluble enough to excrete in urine or bile.

Simple rule to remember: the liver keeps blood chemistry in a narrow, stable range — this stabilizing behavior is called *homeostasis*, and the liver is one of the body's most important homeostatic organs.

Practice problem: After a big pasta dinner, blood glucose rises sharply. Name the process the liver uses to lower it, and explain in one sentence why storing glucose as glycogen is better than just leaving excess glucose in the bloodstream.